

CSCI 4200: Operating System Engineering

Assignment 2 — Software-Based Context Switching with Multitasking

Due Date: March 22, 2026, 23:55

Overview:

In this assignment, you will revise the programs in Assign02-Prog.zip to implement five user-level tasks that print out the characters 'A', 'B', 'C', 'D', and 'E'.

Part I (70 marks)

You are required to add three additional user-level tasks (task2, task3, and task4) to the existing codebase.

- All new tasks must **share the same LDT entry and TSS segment**.
- In boot.s, simply update **TASKNUM** to **five**, so that five tasks (located in Sectors 281–285) can be loaded into physical memory.
- Memory layout:
 - task0 → loaded at 0x50000
 - task1 → loaded at 0x60000
 - task2 → loaded at 0x70000
 - task3 → loaded at 0x80000
 - task4 → loaded at 0x90000
- User-level stack pointers:
 - Each task's user-level stack is initialized to addresses in the range 0x60000–0xA0000, respectively.
- Each task's virtual address space:
 - 0–2G → user space
 - 2G–4G → kernel space

Part II (30 marks)

Replace the LDT with GDT-based segments to run the five tasks.

- Set up two GDT entries: one for the user-level code segment and one for the user-level data segment
- All five tasks must share these GDT entries

Submission Instructions

Submit two zip files, each containing all required files (source code, headers, Makefile, etc.):

- **part1.zip — Implementation for Part I**
- **part2.zip — Implementation for Part II**